LeadFlow – B2B Lead Management & Marketing Campaign Platform

Full-Stack Development Intern | Adz4needz

Project Assignment Document — Build a real working MVP connecting digital marketing campaigns, lead generation, sales follow-up and customer conversion.

1. Project Objective

Build a professional web-based **B2B Lead Management & Marketing Campaign Platform** for an organization that generates potential customers through digital marketing channels.

LeadFlow should connect the complete journey from marketing activity to lead generation, sales follow-up and customer conversion.

Core objective: **Track every lead from its marketing source to its final outcome.**

2. Technology & Cost Requirements

Use free/open-source technologies and free tiers wherever possible.

Possible choices include React/Next.js, TypeScript, Java/Spring Boot, PostgreSQL, Supabase, free authentication, free UI/chart libraries and free storage tiers.

Do not require paid APIs, paid AI services, payment gateways, paid SMS/WhatsApp services or premium plugins.

The application must be responsive on desktop, tablet and mobile browsers.

3. User Roles

A. Digital Marketing Executive

- Create and manage campaigns
- Manage lead sources
- Manage campaign content
- Manage landing-page information
- View campaign leads
- Track campaign performance
- Record campaign expenses
- View marketing analytics

B. Sales Executive

- View assigned leads
- Update lead status

- Add follow-ups and notes
- Qualify leads
- Schedule follow-ups
- Mark leads converted or lost

C. Admin

- Manage users
- View and manage campaigns and leads
- Assign leads
- Manage lead sources
- View organization dashboards
- Export reports
- Manage system settings

4. Main Application Navigation

Digital Marketing

- Dashboard
- Campaigns
- Leads
- Lead Sources
- Content
- Landing Pages
- Analytics
- Reports

Sales

- Dashboard
- My Leads
- Follow-ups
- Converted Leads
- Lost Leads
- Profile

Admin

- Dashboard
- Users
- Campaigns
- Leads
- Sources

- Reports
- Settings

5. Campaign Management

Create campaigns with campaign ID, campaign name, type, objective, start/end dates, target audience, target location, budget, status and description.

Campaign types may include Social Media, Search, Email, Content Marketing, Website, Referral, Event and Other.

Statuses: Planned, Active, Paused, Completed and Cancelled.

6. Lead Management

Each lead should contain Lead ID, name, company, email, phone, location, industry, job role, lead source, campaign, date received, assigned salesperson, status, lead score and notes.

Statuses: New, Contacted, Qualified, Proposal, Negotiation, Converted and Lost.

Provide duplicate-lead warnings and a clear lead details page.

7. Lead Source Tracking

Support sources such as Google Search, Instagram, Facebook, LinkedIn, YouTube, Website, Email, Referral, Event, Direct and Other.

Allow marketing users to compare lead volume and conversion outcomes by source.

8. Lead Qualification & Assignment

Record company size, requirement, budget range, expected purchase timeline, decision-maker status, business need and qualification status.

Qualification levels: Not Qualified, Potential, Qualified and High Priority.

Authorized users should assign leads to sales executives and record assignment date, priority and notes.

9. Follow-Up Management

Sales users should create follow-ups with date, type, notes, outcome and next follow-up date.

Types: Phone, Email, Meeting, Demo, Online Meeting and Other.

Outcomes: Interested, Follow-up Required, Not Interested, No Response, Converted and Lost.

10. Lead Timeline

Display a chronological timeline such as Lead Created → Assigned → Contacted → Follow-up → Qualified → Demo → Proposal → Converted.

Each event should store event name, date/time, responsible user and notes where applicable.

11. Marketing Content Management

Allow marketing users to manage Social Media Posts, Blogs, Advertisements, Emails, Videos, Landing Pages and Other content.

Record title, campaign, content type, platform, target audience, publication date, status and description.

Statuses: Draft, Scheduled, Published and Archived.

12. Landing Page Management

Record landing page name, campaign, URL, target audience, objective, CTA, leads generated, conversions and status.

The MVP only needs landing-page tracking; a website-builder is not required.

13. Campaign Performance

Record impressions, reach, clicks, leads, qualified leads, conversions, campaign cost and revenue generated.

Where possible, calculate CTR, lead conversion rate, cost per lead, cost per conversion and ROI.

Handle zero-denominator cases safely and display metrics clearly.

14. Dashboards & Analytics

Digital Marketing Dashboard

- Active Campaigns
- Total Leads
- Qualified Leads
- Converted Leads
- Conversion Rate
- Total Campaign Spend
- Cost Per Lead
- Cost Per Conversion
- Leads by Source
- Leads by Campaign
- Conversion by Source
- Monthly Lead Trend

Sales Dashboard

- My Leads
- New Leads
- Follow-ups Due
- Overdue Follow-ups
- Qualified Leads
- Proposals

- Converted Leads
- Lost Leads

Admin Dashboard

- Total campaigns
- Total leads
- Active leads
- Converted leads
- Campaign spend
- Conversion performance
- Lead-source performance

15. Campaign Budget & Expenses

Record campaign expenses with campaign, expense type, amount, date and description.

Expense types: Advertising, Content, Design, Event, Influencer, Software and Other.

Show total campaign expenditure.

16. Customer Conversion & Lost Lead Analysis

On conversion, record conversion date, customer/company, deal value, campaign, source, salesperson and notes.

When a lead is lost, require a reason such as Budget, Competitor, No Requirement, No Response, Timing, Product Mismatch, Pricing or Other.

Preserve the original lead history after conversion or loss.

17. Search, Filters, Reports & Export

Search/filter by lead name, company, campaign, source, salesperson, status, qualification, location, industry and date.

Generate campaign performance, lead generation, source, conversion, expenditure, salesperson and ROI reports.

Allow CSV or Excel-compatible export.

18. Notifications

Provide in-app notifications for new lead assignment, follow-up due/overdue, conversion, lost lead, campaign ending and campaign completion.

No paid SMS or WhatsApp integration is required.

19. Validation, Security & Privacy

Validate required fields, email/phone formats, duplicate leads, campaign dates, lead assignments, conversion amounts and follow-up dates.

Require authentication and role-based access.

Users should only access information permitted by their role.

Important lead and campaign changes should be traceable.

Do not expose lead information publicly.

20. Suggested Database Entities

Possible entities: Users, Campaigns, Campaign Expenses, Lead Sources, Leads, Lead Assignments, Lead Activities, Follow-ups, Marketing Content, Landing Pages, Lead Qualification, Conversions, Lost Lead Reasons and Notifications.

The intern should determine the final database structure and relationships.

21. UI/UX Requirements

Professional, modern, clean and responsive interface.

Use dashboard cards, tables, status badges, search, filters, charts, forms, lead timelines, confirmation dialogs, loading states, empty states and error states.

The lead details page should provide a complete view of the lead journey.

22. Product Boundaries & Future Roadmap

Do not add AI lead prediction, AI-generated marketing content, automated customer decisions, paid advertising APIs, WhatsApp API, SMS API or payment gateway in the MVP.

Future phases may include website lead-form integration, email campaign integration, social media integrations, advanced attribution, CRM functionality, marketing automation and AI-assisted analytics.

Final Requirement

Build a fully functional MVP that allows the organization to create campaigns, manage sources and leads, assign and qualify leads, manage follow-ups, track marketing content and landing pages, record campaign spending, convert or close leads, view complete timelines, monitor dashboards, search/filter data, generate reports, export data, receive notifications and enforce role-based access. The application must be a working product, not only a UI prototype. The intern is responsible for selecting the architecture, technologies and implementation approach.